

1	Course Name:
	Machine Learning
2	Course Code:
	IIB20203
3	Year and Semester Offered:
	Year 2 Semester 2
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	IGB20303 Probability and Statistics for IT
6	Course Synopsis:
	With the increased availability of data from varied sources there has been increasing attention paid to the various data driven disciplines such as analytics and machine learning. In this course we intend to introduce some of the basic concepts of machine learning from a mathematically well motivated perspective. We will cover the different learning paradigms and some of the more popular algorithms and architectures used in each of these paradigms.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Understand a wide variety of learning algorithms. 2. Apply the algorithms to a real-world problem. 3. Optimize the models learned and report on the expected accuracy that can be achieved by applying the models.
8	Content Outline:
	Topic
	1. Introductory Topics
	2. Linear Regression and Feature Selection
	3. Linear Classification
	4. Support Vector Machines and Artificial Neural Networks
	5. Bayesian Learning and Decision Trees
	6. Ensemble Methods
	7. Unsupervised Learning and Clustering